

Advances in Neural Retrieval Systems — Research Paper

Abstract

This paper examines retrieval-augmented generation (RAG) architectures for domain-specific question answering. V

Introduction

Large language models excel at general reasoning but struggle with proprietary or time-sensitive knowledge. RAG a

Methodology

We evaluated three chunking strategies:

1. Fixed-size sliding window (500 tokens, 100 overlap)
2. Recursive character splitting (1000 chars, 200 overlap)
3. Semantic paragraph splitting

Embedding models tested: text-embedding-004, all-MiniLM-L6-v2.

Generation model: Gemini 2.0 Flash.

Top-k retrieval values ranged from 3 to 10. Optimal performance was observed at k=5 with recursive character splitting.

Results & Conclusion

Recursive character splitting with $k=5$ achieved 87.3% answer accuracy on the proprietary document benchmark, outperforming baseline methods.

Key finding: chunk overlap of 150-200 characters significantly reduces boundary truncation errors. Strict grounding protocol improved precision by 12%.

Future work includes multi-hop retrieval and citation verification pipelines.